# Zain Ul Abidin

Gen AI & ML Engineer | AI Agents, RAG Systems & Machine Learning

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## PROFESSIONAL SUMMARY

Software Engineering student and Gen AI & ML engineer who builds LLM agents, RAG chatbots and computer-vision models from idea to working deployment. Currently interning at Excellence Code Solution, working on real-world RAG systems and AI agents. Strong focus on clean APIs, fast responses and honest evaluation. Open to on-site, hybrid or remote roles.

## EXPERIENCE

**Gen AI / ML Engineer (Intern), Excellence Code Solution, Lahore, Pakistan | Sep 2026 - Present**

- Building real-world projects with RAG systems and AI agents.
- Trained in Generative AI and applied it to end-to-end projects deployed live.

## PROJECTS

**1. Leave Management AI Agent** (Python, FastAPI, LangChain, LangGraph, Groq, Supabase, Resend, Render, Vercel)
- Conversational agent that checks an employee's leave balance, confirms eligibility, files the request and emails the manager one-click Approve/Reject links, balance and confirmation email update automatically.
- Handled approval state safely, single-use email links and conversation memory; deployed end-to-end to production.

**2. Movie Dekh - Movie Recommendation Engine with AI Chatbot "Moodey"** (NLP, Pandas, FastAPI, Groq, Serper, TMDB API)
- NLP-based personalized movie recommendations plus an LLM agent that finds movies from scene descriptions and answers movie questions.

**3. Helmet Detection** (TensorFlow, Keras, OpenCV, MobileNetV3)
- Real-time helmet detection on live webcam feed; improved accuracy from about 70% (CNN from scratch)
- to 95% using transfer learning (MobileNetV2 86%, EfficientNetB0 90%, MobileNetV3 95%).

**4. Palm Tree Counter from Drone Images** (YOLO, Ultralytics, Roboflow, OpenCV)
- Object detection system that detects and counts palm trees in aerial drone video using transfer learning.

**5. Restaurant RAG Chatbot** (RAG, LangChain, FastAPI, Python)
- RAG-based chatbot that answers restaurant questions and helps customers choose from the menu.

## TECHNICAL SKILLS

- **Gen AI:** RAG, AI Agents, LLMs, LangChain, LangGraph, Prompt Engineering, Groq
- **ML & Vision:** Machine Learning, Scikit-learn, TensorFlow, Keras, Computer Vision, OpenCV, YOLO, Transfer Learning, NLP
- **Languages & Data:** Python, C++, Pandas, Matplotlib
- **Backend & Deployment:** FastAPI, Supabase (Postgres), Render, Vercel
- **Other:** Graphic Design

## EDUCATION

- BS Computer Software Engineering, Virtual University (2024 - Present)
- AI Engineering, PNY Trainings (Apr 2026 - Oct 2026)
- AI Prompt Engineering, Outskill (Dec 2025)
- ICS Computer Science, KIPS Education System (from 2022)
- Graphic Design, Expert B Computers (Oct 2021 - Dec 2021)

## CERTIFICATIONS

- RAG Engineering: Build AI That Knows Your Data - Beyond Tahir (Oct 2026)
- Introduction to Generative AI - Beyond Tahir (Oct 2026)
- Generative AI Mastermind - Outskill
- C++ Essentials 1 - Cisco Networking Academy via Virtual University (May 2025)